

Machine Learning (ML) Lab Project

The ML lab project contains the following files:

- The lab skeleton, a C++ project.
- Iris dataset
- Boston Housing price dataset

1. The Lab Skeleton

All the labs will be conducted using the Lab skeleton developed using C++. The Lab skeleton provides the visual interface, as shown in Figure 1.

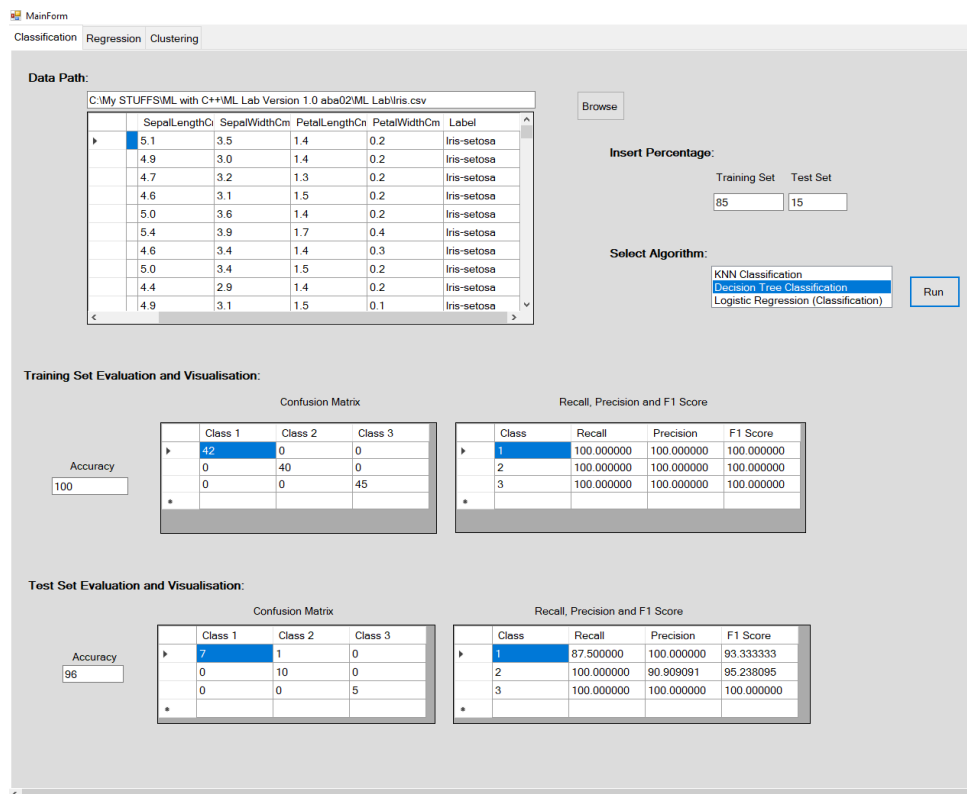


Figure 1: Example of Classification Interface after implementing and running Decision Tree Algorithm.

The interface has the following functionality:

1. Data loading
2. Three Tabs for Classification, Regression and Clustering tasks
3. Training and Test data split
4. Option for algorithm selection
5. Visualization of results

For the Labs, the above functionalities (i.e., C++ code) related to the visual interface are provided with the skeleton. In the labs, you will implement the algorithms from scratch in the corresponding C++ classes and demonstrate the results to the lab assistants.

The file structure of the ML C++ project is shown in Figure 2. The C++ Class files are divided into several groups. Classes under *DataUtils* and *Evaluations* are already implemented for the labs. The *PCADimensionalityReduction.cpp* class under *Utils* has already been implemented, which you will need to use in the Clustering lab. In the labs, your task will primarily be to implement the **training** and **prediction** functions of the machine learning algorithms and use the related functions to visualize the results.

The project also uses an external Library called *Eigen*, a template library for Linear Algebra.

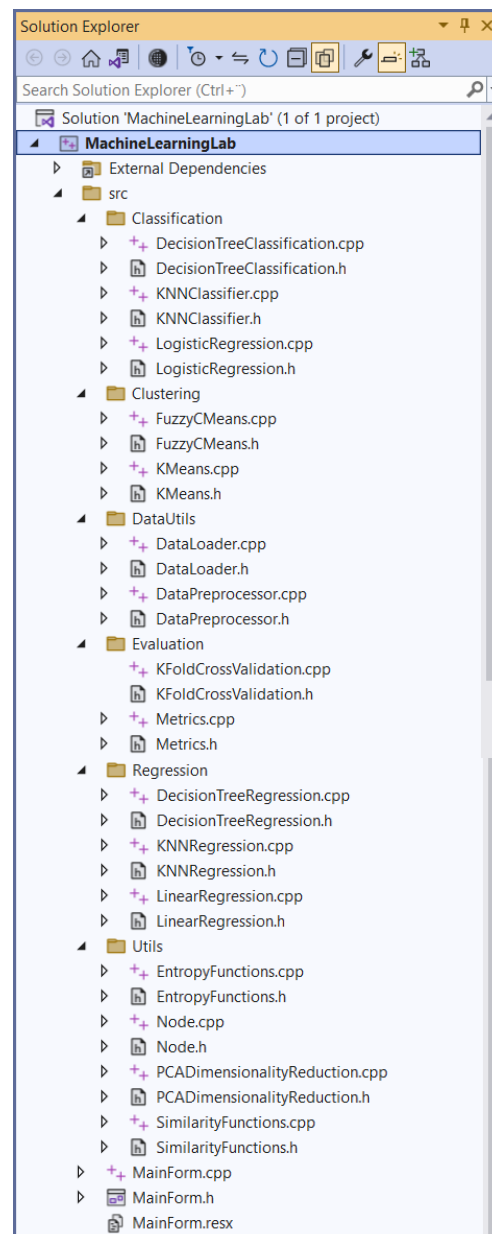


Figure 2: ML Project File Structure

1.1. Eigen template library for Linear Algebra

We are using the Eigen C++ library for linear algebra computation, and it is an additional library that is required for the Lab skeleton.

Eigen is a C++ template library for linear algebra: matrices, vectors, numerical solvers, and related algorithms.

- Here is a link to the Eigen library website:
https://eigen.tuxfamily.org/index.php?title=Main_Page.
- Download: The Eigen library can be downloaded from the following link:
<https://gitlab.com/libeigen/eigen/-/archive/3.4.0/eigen-3.4.0.zip>.
- You can find tutorial on Eigen at:
<https://eigen.tuxfamily.org/dox/GettingStarted.html>

1.2. Eigen library configuration

The Eigen library is included with the Lab skeleton, but depending on where you have stored files in your computer it may require reconfiguration. Take the following steps to configure the Eigen library.

1. After downloading, extract the project zip file.
2. Open the project with Microsoft Visual Studio.
3. In Visual Studio you can configure the Eigen (see Figure 3) as follows:
 1. Go to the project name and press right-click.
 2. Select properties.
 3. In the C/C++ tab, choose General.
 4. In the first option, “Additional Include Directories”, put the path of the Eigen library.
 5. Press Apply the configuration and OK.

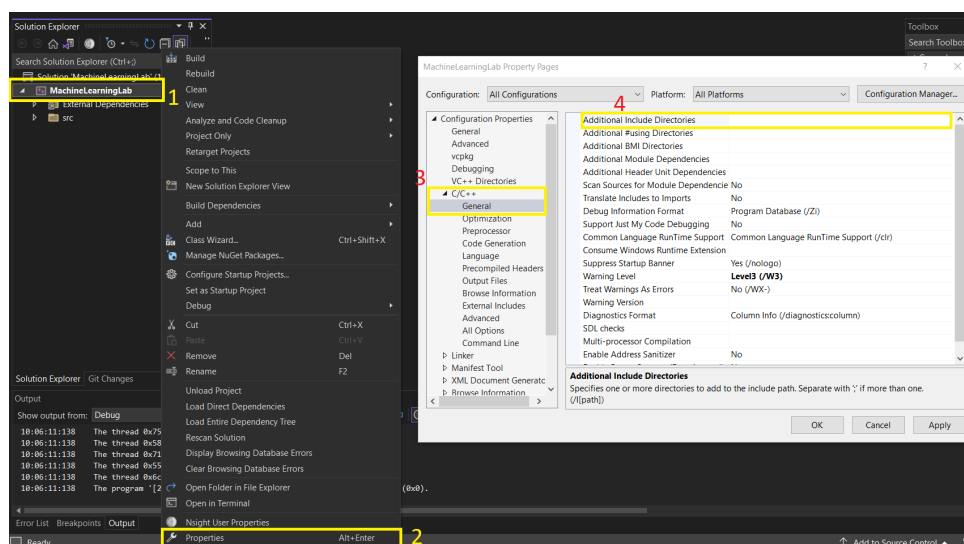


Figure 3: Eigen Library Configuration

Note: If there is any problem with the Eigen library, please download a fresh copy from the link above.

2. Dataset Description

In the labs we are going to use two well-known benchmark datasets: **Iris** dataset and **Boston Housing Price** dataset. The Iris dataset will be used for *Classification* and *Clustering* problems, and the Boston Housing price will be used for *Regression* problems. The descriptions of the datasets are given bellow.

2.1. Iris dataset

The Iris data set is the most well-known in pattern recognition. The data set contains three classes of 50 instances each, where each class refers to a type of iris plant. One class is linearly separable from the other two categories; the latter are not linearly separable from each other.

The summary of the dataset:

1. Number of Instances: 150 (50 in each of three classes)
2. Number of Attributes: 4 numeric, predictive attributes and the class
3. Attribute Information:
 - a. sepal length in cm
 - b. sepal width in cm
 - c. petal length in cm
 - d. petal width in cm
4. Target Class:
 - a. Iris Setosa
 - b. Iris Versicolour
 - c. Iris Virginica
5. There are no missing values for any attribute.

Summary Statistics:

	Min	Max	Mean	SD	Class Correlation
Sepal length	4.3	7.9	5.84	0.83	0.7826
Sepal width	2.0	4.4	3.05	0.43	- 0.4194
Petal length	1.0	6.9	3.76	1.76	0.9490 (high!)
Petal width	0.1	2.5	1.20	0.76	0.9565 (high!)

Class Distribution: 33.3% for each of the three classes.

2.2. Boston Housing Price dataset

The data set contains information about different houses in Boston. This data was originally a part of the UCI Machine Learning Repository and has been removed now. There are 506 samples and 13 feature variables in this dataset. The objective is to predict the value of house prices using the given features.

The summary of the dataset:

1. Attribute Characteristics: Multivariate, i.e., Categorical, Integer, Real
2. Number of Instances: 506
3. Number of Attributes: 14 Date Donated 19930707
4. Associated Tasks: Regression
5. There are no missing values for any attribute.
6. Attribute Information:
 - a. CRIM: per capita crime rate by town
 - b. ZN: proportion of residential land zoned for lots over 25,000 sq.ft.
 - c. INDUS: proportion of nonretail business acres per town
 - d. CHAS: Charles River dummy variable (= 1 if tract bounds river; 0 otherwise)
 - e. NOX: nitric oxides concentration (parts per 10 million)
 - f. RM: average number of rooms per dwelling
 - g. AGE: proportion of owner occupied units built prior to 1940
 - h. DIS: weighted distances to five Boston employment centres
 - i. RAD: index of accessibility to radial highways
 - j. TAX: full value property tax rate per \$10,000
 - k. PTRATIO: pupil teacher ratio by town
 - l. B: $1000(B_k - 0.63)^2$ where B_k is the proportion of blacks by town
 - m. LSTAT: % lower status of the population
7. Target Class
 - a. MEDV: Median value of owner-occupied homes in \$1000's

Figure 4 shows the distribution of the target variable MEDV, and we can see that the values of MEDV are distributed normally with few outliers.

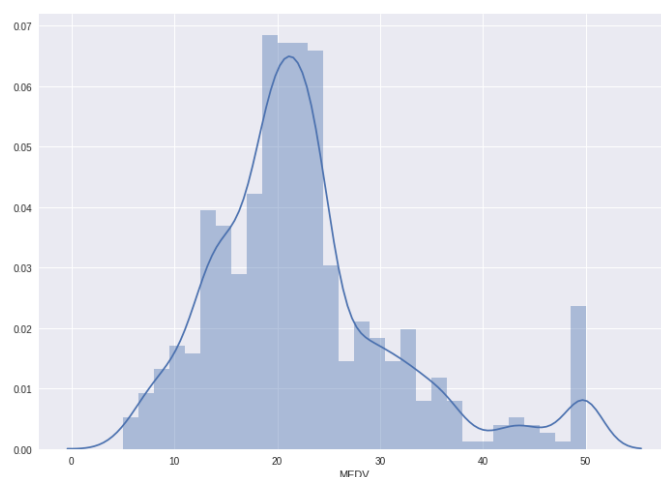


Figure 4: Distribution of the target variable MEDV